Sections: B,C & D

Linear Algebra (MT-1004)

Homework#28

Instructor: Muhammad Ali

Polynomial Regression

Question 1

A real estate company wants to predict the price of houses (in thousands) based on three factors: the size of the house (in square feet), the number of bedrooms and the number of bathrooms. The dataset is as follows:

Size (Square Feet)	Bedrooms	Bathrooms	Price (Y in thousands)
2000	3	2	400
1500	2	1	250
3000	4	3	600
1800	3	2	350
2500	4	2	500

- Write the design matrix for given data
- Compute the coefficients b (Bias) and w_1 (Weights)
- Perform a quadratic regression model on the provided dataset
- Predict the price of a house when the size is 2300 square feet, the number of bedrooms is 3 and the number of bathrooms is 2